Full solution packet for arXiv:2108.02727

Collapsing subsets in Lipschitz-free spaces and an explicit flux model for persistence diagrams

Status. full__solution__likely__valid. This packet answers the concrete problem in the conclusion of Chad Giusti and Darrick Lee, *Signatures, Lipschitz-free spaces, and paths of persistence diagrams*, arXiv:2108.02727. It first gives the exact functorial answer for every metric pair, then combines it with the Euclidean representation of Cúth–Kalenda–Kaplický to obtain an explicit vector-field representation for ordinary persistence diagrams.

Source question (Conclusion, p. 26). The source asks for "an explicit description of the Lipschitz-free space of the quotient metric space of a metric pair" and, specifically, whether known representations of $\mathcal{F}(\mathbb{R}^N)$ can be adapted to represent persistence diagrams.

> - **Lipschitz-free spaces.** Although we have discussed several properties of Lipschitz-free spaces which are directly applicable to the study of persistence diagrams, are there other aspects of this connection which we can exploit? One concrete problem is the an explicit description of the Lipschitz-free space of the quotient metric space of a metric pair (X, d, A). Can we adapt the representations for known Lipschitz-free spaces such as $\mathcal{F}(\mathbb{R}^N)$ [29] to develop a new representation of persistence diagrams?

Result at a glance

Let X/A denote the pointed metric space obtained by collapsing a nonempty closed subset A of a metric space X to one point. With X based at a point of A, put

$$\mathcal{F}_X(A) = \overline{\operatorname{span}}\{\delta_X(a) : a \in A\} \subseteq \mathcal{F}(X).$$

There is a canonical linear isometry

$$\boxed{\mathcal{F}(X/A) \cong \mathcal{F}(X)/\mathcal{F}_X(A).} \tag{1}$$

Thus every concrete model of $\mathcal{F}(X)$ automatically induces one for the collapsed metric pair: quotient the model by the vectors representing A.

For ordinary persistence diagrams, rotate the birth–death half-plane so that

$$H = \mathbb{R} \times (0, \infty), \qquad \partial H = \mathbb{R} \times \{0\}.$$

The diagonal becomes ∂H. Then (1) and the theorem of Cúth–Kalenda–Kaplický give the explicit isometric representation

$$\boxed{\mathcal{F}(\overline{H}/\partial H) \cong \frac{L^1(H; \mathbb{R}^2)}{\{g \in L^1(H; \mathbb{R}^2) : \operatorname{div} g = 0 \text{ in } \mathcal{D}'(H)\}}.} \tag{2}$$

The fact that divergence is taken *inside* H, rather than after zero-extension to $\mathbb{R}^2$, is the key boundary effect: flux may disappear through the collapsed diagonal.

If D, E are finite persistence diagrams and ν_D, ν_E are their counting measures after rotation into H, then (2) gives the exact flux formula

$$\boxed{W_1^\partial(D, E) = \inf\left\{\|g\|_{L^1(H)} : g \in L^1(H; \mathbb{R}^2),\ \operatorname{div} g = \nu_E - \nu_D \text{ in } \mathcal{D}'(H)\right\}.} \tag{3}$$

Proof intuition

There are two simple dual facts behind the result.

First, a Lipschitz function on X/A is exactly a Lipschitz function on X that vanishes on A. The two Lipschitz constants agree: besides the direct bound through $d(x,y)$, vanishing on A gives the second bound through $d(x,A)+d(y,A)$. Since Lipschitz-free spaces are the canonical preduals of these function spaces, collapsing A amounts to quotienting out the closed span of the evaluation vectors on A.

Second, in the Euclidean representation a vector field g acts on a Lipschitz function f by $\int g \cdot \nabla f$. After the boundary is collapsed, the only test functions are those with zero boundary values. Consequently, the invisible vector fields are precisely those whose divergence vanishes in the interior. Boundary flux is deliberately not tested. An atom of a persistence diagram is a unit source whose flux can either travel to another atom or exit through the boundary; minimizing the L^1 cost gives exactly partial W_1.

The general metric-pair theorem

For a nonempty closed $A \subseteq X$, write

$$d_A([x],[y]) = \min\{d(x,y), d(x,A)+d(y,A)\}$$

for the quotient metric, with the evident interpretation when one class is the collapsed point $*$. Choose $a_0 \in A$ as the base point of X.

Theorem 1 (Free spaces commute with collapsing)**.** *Let X be a metric space and let $A \subseteq X$ be nonempty and closed. The quotient map $q : X \to X/A$ induces a canonical surjective linear map*

$$Q : \mathcal{F}(X) \longrightarrow \mathcal{F}(X/A), \qquad Q\delta_X(x) = \delta_{X/A}(q(x)),$$

and

$$\ker Q = \mathcal{F}_X(A).$$

The induced map $\mathcal{F}(X)/\mathcal{F}_X(A) \to \mathcal{F}(X/A)$ is a linear isometry.

Proof. Let

$$I_A = \{f \in \mathrm{Lip}_0(X) : f|_A = 0\}.$$

Composition with q maps $\mathrm{Lip}_0(X/A)$ into I_A. This map is onto: if $f \in I_A$, define $\widetilde{f}([x]) = f(x)$ and $\widetilde{f}(*) = 0$. It is also isometric. Indeed, if $L = \|f\|_{\mathrm{Lip}}$, then

$$|f(x) - f(y)| \le L d(x,y)$$

and, because f vanishes on A,

$$|f(x) - f(y)| \le |f(x)| + |f(y)| \le L\big(d(x,A) + d(y,A)\big).$$

Taking the smaller bound shows that $\widetilde{f}$ is L-Lipschitz on X/A. The reverse norm inequality follows because q is 1-Lipschitz. Therefore

$$\mathrm{Lip}_0(X/A) \xrightarrow[cong]{f \mapsto f\circ q} I_A. \tag{4}$$

The annihilator of $\mathcal{F}_X(A)$ in $\mathcal{F}(X)^* = \mathrm{Lip}_0(X)$ is exactly I_A. The adjoint of the induced map

$$\overline{Q} : \mathcal{F}(X)/\mathcal{F}_X(A) \longrightarrow \mathcal{F}(X/A)$$

is precisely (4), hence is an onto isometry. It follows directly that $\overline{Q}$ is an isometry: for $u \in \mathcal{F}(X)/\mathcal{F}_X(A)$,

$$\|\overline{Q}u\| = \sup_{\|h\|_{\mathrm{Lip}} \le 1} |\langle u, h \circ q \rangle| = \|u\|.$$

Its range is closed, and its annihilator is $\ker \overline{Q}^* = \{0\}$, so the range is also dense. Thus $\overline{Q}$ is onto. The asserted kernel identity follows. □

Interpretation. The theorem is base-point independent in the usual canonical sense. The choice $a_0 \in A$ merely makes every $\delta_X(a)$ unambiguous. Also, $\mathcal{F}_X(A)$ is canonically the embedded copy of $\mathcal{F}(A)$; the isometry of that embedding follows, for example, by the scalar McShane extension theorem.

Euclidean metric pairs

We record the exact mechanism for adapting the known Euclidean model. Let $U \subset \mathbb{R}^n$ be nonempty, open, and convex, and let A be a nonempty closed subset of $\overline{U}$. Define

$$\mathcal{Y}_A = \left\{ g \in L^1(U;\mathbb{R}^n) : \int_U g \cdot \nabla f = 0 \quad \text{for every } f \in \mathrm{Lip}(\overline{U}) \text{ with } f|_A = 0 \right\}. \tag{5}$$

Proposition 2 (Explicit model for a Euclidean pair)**.** *There is a canonical linear isometry*

$$\mathcal{F}(\overline{U}/A) \cong L^1(U;\mathbb{R}^n)/\mathcal{Y}_A. \tag{6}$$

Under (6), g acts on $f \in \mathrm{Lip}(\overline{U})$ with $f|_A = 0$ by $\int_U g \cdot \nabla f$.

Proof. Cúth–Kalenda–Kaplický, Theorem 1.1, identify $\mathcal{F}(U)$ isometrically with a quotient of $L^1(U;\mathbb{R}^n)$ via the pairing

$$(g, f) \longmapsto \int_U g \cdot \nabla f. \tag{7}$$

Passing from U to its completion $\overline{U}$ does not change the free space: the evaluation vectors from the dense subset U have dense span and the metric embedding extends continuously. Changing the base point only adds constants to dual Lipschitz functions, so it does not alter the gradient pairing (7). Apply Theorem 1. By (4), the dual functions that remain after collapsing A are exactly the Lipschitz functions vanishing on A. Therefore the preimage, under (7), of the subspace being quotiented out is exactly the annihilator (5). Taking the quotient gives (6), with no change of norm. □

For comparison, before collapsing a boundary the kernel in the Cúth–Kalenda–Kaplický theorem consists of fields whose zero-extension has divergence zero on all of $\mathbb{R}^n$. Formula (5) shows exactly how a collapsed subset changes that condition.

The persistence half-plane

Let $H = \mathbb{R} \times (0, \infty)$ and $A = \partial H$. Distributional divergence in H is defined by

$$\langle \operatorname{div} g, \varphi \rangle = -\int_H g \cdot \nabla\varphi \qquad (\varphi \in C_c^\infty(H)).$$

Lemma 3 (The collapsed-boundary kernel)**.** *For the pair $(\overline{H}, \partial H)$, the subspace in (5) is*

$$\mathcal{Y}_{\partial H} = \mathcal{Z}_H := \{g \in L^1(H; \mathbb{R}^2) : \operatorname{div} g = 0 \text{ in } \mathcal{D}'(H)\}. \tag{8}$$

Proof. If $g \in \mathcal{Y}_{\partial H}$, test (5) with $\varphi \in C_c^\infty(H)$ to obtain $\operatorname{div} g = 0$ in H.

Conversely, suppose $\operatorname{div} g = 0$ in H, and let $f \in \operatorname{Lip}(\overline{H})$ vanish on ∂H. Extend f by zero to the lower half-plane. The extension F is Lipschitz on $\mathbb{R}^2$: for $(s,t) \in H$, $|f(s,t)| \le \|f\|_{\mathrm{Lip}} t$, which also handles pairs lying on opposite sides of the boundary.

Choose $\chi_R \in C_c^\infty(\mathbb{R}^2)$ with $\chi_R = 1$ on $B(0,R)$, $\chi_R = 0$ outside $B(0, 2R)$, and $|\nabla\chi_R| \le C/R$. Put $F_R = \chi_R F$. Translate F_R upward by ε and mollify at a scale smaller than $\varepsilon/2$. This produces functions in $C_c^\infty(H)$ whose gradients converge almost everywhere to ∇F_R and are bounded uniformly in L^∞. Pairing with $g \in L^1$ and using dominated convergence gives

$$\int_H g \cdot \nabla F_R = 0. \tag{9}$$

As $R \to \infty$, $\nabla F_R \to \nabla f$ almost everywhere on H. Moreover the gradients are uniformly bounded: on the annulus where $\nabla\chi_R \neq 0$,

$$|F|\,|\nabla\chi_R| \le \|f\|_{\mathrm{Lip}} t\, C/R \le 2C\|f\|_{\mathrm{Lip}}.$$

Dominated convergence in (9) now yields $\int_H g \cdot \nabla f = 0$. Thus $g \in \mathcal{Y}_{\partial H}$. □

Corollary 4 (Vector-field model for persistence diagrams)**.** *Let*

$$\Omega = \{(b,d) \in \mathbb{R}^2 : b \le d\}, \qquad \Delta = \{(t,t) : t \in \mathbb{R}\},$$

with the Euclidean metric. Then

$$\mathcal{F}(\Omega/\Delta) \cong L^1(H; \mathbb{R}^2)/\mathcal{Z}_H \tag{10}$$

canonically and isometrically, after the rotation

$$\rho(b,d) = \left(\frac{b+d}{\sqrt{2}}, \frac{d-b}{\sqrt{2}}\right).$$

For finite diagrams D, E, let $\nu_D = \sum_{z \in D} \delta_{\rho(z)}$ and define ν_E similarly, omitting diagonal points. The class of $D - E$ consists exactly of the fields g with

$$\operatorname{div} g = \nu_E - \nu_D \quad \text{in } \mathcal{D}'(H), \tag{11}$$

and its norm is the infimum of $\|g\|_{L^1(H)}$ over (11). Equivalently, formula (3) holds.

Proof. The rotation ρ is an isometry from Ω onto $\overline{H}$ and maps Δ onto ∂H. Proposition 2 and Lemma 3 give (10).

The pairing convention gives

$$\int_H g \cdot \nabla f = -\langle \operatorname{div} g, f \rangle.$$

Hence (11) represents $\langle \nu_D - \nu_E, f \rangle$, exactly the evaluation vector of $D - E$. Two representatives differ precisely by an element of $\mathcal{Z}_H$, and the quotient norm is the stated infimum. Finally, Giusti–Lee define $\mathcal{F}(\Omega/\Delta)$ as the completion of finite virtual diagrams in the partial W_1 norm. Therefore that quotient norm equals $W_1^\partial(D, E)$. □

For a single pair $x, y \in H$, Corollary 4 specializes to the transparent identity

$$\inf_{\operatorname{div} g=\delta_y-\delta_x} \|g\|_1 = \min\{|x-y|, \operatorname{dist}(x,\partial H)+\operatorname{dist}(y,\partial H)\}.$$

The two terms are the two possible flux routes: directly from x to y, or from x to the collapsed boundary and back out to y.

Verification and scope

Checks performed. The quotient metric used in Theorem 1 was checked against the source definition. The norm equality in (4) uses both branches of that minimum, so it is not merely a contractive identification. The supporting paper's Theorem 1.1 was checked directly, including its warning that its uncollapsed kernel uses zero-extension and divergence in all of $\mathbb{R}^n$. Lemma 3 independently derives the changed boundary condition. The signs in (11) follow from $\langle \operatorname{div} g, f\rangle = -\int g \cdot \nabla f$.

What "explicit" means here. Formula (10) is an exact, isometric Banach-space quotient, and (11) gives a concrete PDE description of every diagram vector. It is not a unique coordinate representation: adding a divergence-free field does not change the element. Nor is it finite-dimensional. Those limitations already occur in the known representation of $\mathcal{F}(\mathbb{R}^2)$ that motivated the source question.

Novelty assessment. Exact-phrase and structural searches were run for the quotient identity, a collapsed-subset representation, and a divergence/flux representation of persistence diagrams. The searches recovered the ambient convex-domain theorem used above and later extensions of ambient free-space representations, but no explicit collapsed-diagonal formula (10) or diagram flux formula (3). The abstract identity (1) is elementary enough that it may be folklore; the claimed mathematical contribution of this packet is the complete derivation and its persistence-diagram specialization. Novelty confidence is therefore moderate, while proof confidence is high.

Human-review recommendation. Review as a full answer to the concrete problem on p. 26. The two points most worth checking are the weak-$*$ approximation in Lemma 3 and the interpretation of partial W_1 in Corollary 4. The result concerns the $p = 1$ geometry used by Lipschitz-free spaces; it does not claim analogous linearizations for $p > 1$.

References

[1] C. Giusti and D. Lee, *Signatures, Lipschitz-free spaces, and paths of persistence diagrams*, arXiv:2108.02727, especially the Conclusion, p. 26.

[2] M. Cúth, O. F. K. Kalenda, and P. Kaplický, *Isometric representation of Lipschitz-free spaces over convex domains in finite-dimensional spaces*, Mathematika 63 (2017), 538–552; arXiv:1610.03966, Theorem 1.1.